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BIT 2223 Mobile Computing

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 6

Data transmission in wireless cellular network systems

Objectives

By the end of this topic you should be able to;

- *Understand the concept of cellular networks*
- *Understand the concept of Data transmission in wireless cellular network systems*
- *Understand the concept of circuit switched data services*



6.1. Cellular Networks

Introduction

Mobile communication is the fastest growing field in the telecommunications industry. We look at the history, present state, and future of cellular radio networks. The main objectives are:

- Understand the basic structure of a GSM Network
- Understand the main components of a GSM Network
- Understand advantages of GSM over analogue systems
- Understand how calls are made and received on a GSM network



6.1.1. History of Cellular Radio Networks

In 1946, the first car-based telephone was set up in St. Louis, Missouri, USA. The system used a single radio transmitter on top of a tall building. A single channel was used, therefore requiring a button to be pushed to talk, and released to listen. This half duplex system is still used by modern day CB radio systems utilized by police and taxi operators. In the 1960s, the system was improved to a two-channel system, called the improved mobile telephone system (IMTS). Since frequencies were limited, the system could not support many users.

Cellular radio systems, implemented for the first time in the advanced mobile phone system (AMPS), support more users by allowing reuse of frequencies. AMPS is an analogue system, and is part of first generation cellular radio systems.



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In contrast, second generation systems are digital. In the USA, two standards are used for second generation systems: IS-95 (CDMA) and IS-136 (D-AMPS). Europe consolidated to one system called the global system for mobile communications (GSM). Japan uses a system called personal digital cellular (PDC).

Cellular radio is the fastest growing segment of the communications industry. Because of Europe's early commitment to one system, it is leading the field in both its subscriber base and data transmission capabilities. GSM is used in over 100 countries by over 215 operators inside and outside of Europe. The Japanese PDC system is the second largest digital cellular system, followed by the IS-54/136 and IS-95 systems used in North America.

Current cellular radio systems are in their third generation



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of cellular systems (3G systems) that allows different systems to inter-operate in order to attain global roaming across different cellular radio networks. The International Telecommunication Union (ITU) has been doing research on 3G systems since the mid 1980s. Their version of a 3G system is called international mobile telecommunications - 2000 (IMT-2000). (Though mobile service providers in Kenya are at 2.5 G, they are rolling out 3 G). 3G systems have the following major objectives:

- Use of common global frequencies for all cellular networks.
- Worldwide roaming.
- Standardization of radio interfaces.
- High data transmission rates for both circuit and packet switched data.



- Efficient spectrum utilization schemes.

6.1.2. How it works

A cellular telephone or any other device that can connect to a cellular radio network will be referred to as a mobile station. This is in keeping with the literature on the subject.

A cellular network consists of both land and radio based sections. Such a network is commonly referred to as a PLMN - public land mobile network. The network is composed of the following entities:

- Mobile station (MS): A device used to communicate over the cellular network.
- Base station transceiver (BTS): A transmitter/receiver used to transmit/receive signals over the radio interface section



of the network.

- Mobile switching center (MSC): The heart of the network which sets up and maintains calls made over the network.
- Base station controller (BSC): Controls communication between a group of BTSs and a single MSC.
- Public switched telephone network (PSTN): The land based section of the network.

The figure below illustrates how these entities are related to one another within the network. The BTSs and their controlling BSC are often collectively referred to as the base station subsystem (BSS). The cellular topology of the network is a result of limited radio spectrum. In order to use the radio spectrum efficiently, the same frequencies are reused in nonadjacent cells.



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A geographic region is divided up into cells. Each cell has a BTS that transmits data via a radio link to MSC within the cell. A group of BTSs are connected to a BSC. A group of BSCs are in turn connected to a mobile switching center via microwave links or telephone lines. The MSC connects to the public switched telephone network, which switches calls to other mobile stations or land based telephones.

6.2. Mobile IN

All intelligent networking for telecommunications involves the concept of a "query / response" system. This system entails the notion of distributed intelligence wherein a database is queried for information necessary for call processing.

For example, a mobile communication switch or Mobile Switch-



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ing Center (MSC), that is equipped with mobile IN call logic, can launch a message or "query" to a database hosted by a network element called a Service Control Point (SCP). The SCP processes the request and issues a "response" to the MSC so that it may continue call processing as appropriate.

Wireless intelligent networking is more complex than fixed network based intelligent networking due to the dynamic nature of mobile communications and thus the need for mobility management.

As the world evolves to embrace Internet protocol (IP) technologies, the notion of "network intelligence" is also evolving. The need to leverage IN to support convergence of traditional circuit switched with packet-based networks grows daily.



6.2.1. SS7

SS7 is a critical component of modern telecommunications systems. SS7 is a communications protocol that provides signaling and control for various network services and capabilities. While the Internet, wireless data, and related technology have captured the attention of millions, many forget or don't realize the importance of SS7. Every call in every network is dependent on SS7. Likewise, every mobile phone user is dependent on SS7 to allow inter-network roaming. SS7 is also the "glue" that sticks together circuit switched (traditional) networks with Internet protocol based networks.



- SS7 Architecture

SS7 is comprised of a series of interconnected network elements such as switches, databases, and routing nodes. Each of these elements is interconnected with links, each of which has a specific purpose. The routing nodes are the heart of the SS7 network and are called a Signal Transfer Point (STP). STPs are connected to Service Switching Points (SSP) that are switches equipped with SS7 control logic. SSP switches are connected to the STPs via Access links (A links). STPs also connect to databases called Service Control Points (SCP) via A links. The SCP is the network element that contains service control logic such as instructions for converting an 8XX (toll-free) number into a routable number.



6.2.2. WIN, INAP, and CAMEL

Wireless Intelligent Network (WIN), Intelligent Network Application Part (INAP), and Customized Applications for Mobile Enhanced Logic (CAMEL) mobile IN technologies all have their origin in the same Intelligent Network Conceptual Model (INCM). However, WIN is the standard for ANSI-41 based networks, CAMEL is the standard for GSM based networks, and INAP is actually a European fixed network standard for which many vendors have created proprietary extensions to enable service delivery capabilities.

6.2.3. Capabilities

The reasons for deploying mobile IN technologies are to provide value-added capabilities for purposes such as cost reduction, im-



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proved service delivery, increased variety and quality of services, and rapid service creation and deployment.

These capabilities rely on underlying mobile networking technologies to handle mobility management functions. Together with mobile networking, mobile IN capabilities allow the mobile network operator to deploy a variety of advanced and/or value-added applications.

6.2.4. Applications

Mobile IN enables value-added applications such as:

- One number service
- Prepaid service
- Location based services



- Call management services
- FreePhone (toll-free calling)

6.3. Frequency Division Multiple Access

This was the initial multiple-access technique for cellular systems in which each individual user is assigned a pair of frequencies while making or receiving a call. One frequency is used for downlink and one pair for uplink. This is called frequency division duplexing (FDD). That allocated frequency pair is not used in the same cell or adjacent cells during the call so as to reduce the co channel interference. Even though the user may not be talking, the spectrum cannot be reassigned as long as a call is in place. Different users can use the same frequency in the same cell except that they must transmit at different times. The fea-



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tures of FDMA are as follows: The FDMA channel carries only one phone circuit at a time. If an FDMA channel is not in use, then it sits idle and it cannot be used by other users to increase share capacity. After the assignment of the voice channel the BS and the MS transmit simultaneously and continuously. The bandwidths of FDMA systems are generally narrow i.e. FDMA is usually implemented in a narrow band system. The symbol time is large compared to the average delay spread. The complexity of the FDMA mobile systems is lower than that of TDMA mobile systems. FDMA requires tight filtering to minimize the adjacent channel interference.



6.4. Spread Spectrum Multiple Access

Spread spectrum multiple access (SSMA) uses signals which have a transmission bandwidth whose magnitude is greater than the minimum required RF bandwidth. A pseudo noise (PN) sequence converts a narrow band signal to a wide-band noise like signal before transmission. SSMA is not very bandwidth efficient when used by a single user. However since many users can share the same spread spectrum bandwidth without interfering with one another, spread spectrum systems become bandwidth efficient in a multiple user environment.

There are two main types of spread spectrum multiple access techniques:

1. Frequency hopped multiple access (FHMA)



2. Direct sequence multiple access (DSMA) or Code division multiple access (CDMA).

6.4.1. Frequency Hopped Multiple Access (FHMA)

This is a digital multiple access system in which the carrier frequencies of the individual users are varied in a pseudo random fashion within a wide band channel. The digital data is broken into uniform sized bursts which is then transmitted on different carrier frequencies.

6.5. Space Division Multiple Access

SDMA utilizes the spatial separation of the users in order to optimize the use of the frequency spectrum. A primitive form of SDMA is when the same frequency is reused in different cells



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in a cellular wireless network. The radiated power of each user is controlled by Space division multiple access. SDMA serves different users by using spot beam antenna. These areas may be served by the same frequency or different frequencies. However for limited co-channel interference it is required that the cells be sufficiently separated. This limits the number of cells a region can be divided into and hence limits the frequency re-use factor. A more advanced approach can further increase the capacity of the network. This technique would enable frequency re-use within the cell. In a practical cellular environment it is improbable to have just one transmitter fall within the receiver beam width. Therefore it becomes imperative to use other multiple access techniques in conjunction with SDMA. When different areas are covered by the antenna beam, frequency can be re-



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used, in which case TDMA or CDMA is employed, for different frequencies FDMA can be used.

6.6. Time Division Multiple Access

TDMA allows several users to share the same frequency channel by dividing the signal into different time slots. Users transmit in rapid succession, one after the other, each using his own time slot. It allows multiple stations to share the same transmission medium (e.g. radio frequency channel) while using only a part of its channel capacity. In digital systems, continuous transmission is not required because users do not use the allotted bandwidth all the time. In such cases, TDMA is a complimentary access technique to FDMA. Global Systems for Mobile communications (GSM) uses the TDMA technique. In TDMA, the entire



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bandwidth is available to the user but only for a nite period of time. In most cases the available bandwidth is divided into fewer channels compared to FDMA and the users are allotted time slots during which they have the entire channel bandwidth at their disposal. TDMA requires careful time synchronization since users share the bandwidth in the frequency domain. The number of channels are less, inter channel interference is almost negligible. TDMA uses different time slots for transmission and reception. This type of duplexing is referred to as Time division duplexing (TDD).

The features of TDMA includes the following:

- TDMA shares a single carrier frequency with several users where each users makes use of non overlapping time slots. The number of time slots per frame depends on several



factors such as modulation technique, available bandwidth etc.

- Data transmission in TDMA is not continuous but occurs in bursts. This results in low battery consumption since the subscriber transmitter can be turned OFF when not in use. Because of a discontinuous transmission in TDMA the handoff process is much simpler for a subscriber unit, since it is able to listen to other base stations during idle time slots.
- TDMA uses different time slots for transmission and reception thus duplexers are not required. TDMA has an advantage that is possible to allocate different numbers of time slots per frame to different users. Thus bandwidth can be supplied on demand to different users by concate-



nating or reassigning time slot based on priority.

6.6.1. TDMA Format

- *Trail bits* – allow synchronization of transmissions from mobile units
- *Encrypted bits* – encrypted data
- *Stealing bit* - indicates whether block contains data or is "stolen" for urgent control data
- *Training sequence* – used to adapt parameters of receiver to the current path propagation characteristics - strongest signal selected in case of multi-path propagation
- *Guard bits* – used to avoid overlapping with other bursts



6.7. Mobile Internet Communication Technologies

There are number of mobile communication technologies which have been developed ever since from analog days, new technology is released to cope with telephonic industry demands. Currently we have following mobile and internet communication technologies adopted by different mobile companies in different parts of the world.

- GSM
- CDMA
- EDGE
- GPRS
- VoIP



6.7.1. GSM

GSM stands for Global System for Mobile Communications, but you might think that in that case it should really be GSMC. GSM was originally an acronym for the name of the committee that was doing the standardization work – called Groups Special Mobile. GSM based mobile phones are operated on TDMA Systems, in TDMA single radio frequency is offered to users with any interference. GSM is the largest mobile communication technology worldwide, all manufacturers of Mobile phones develop their products based on GSM, and all mobile companies provide their subscribers GSM networks. GSM standards for Global System for Mobile Communication, it is widely used mobile technology worldwide and it adopted by more than 214 countries around the world.



• GSM Network Architecture

- Mobile station
- Base Station Subsystem
- Network Subsystem
- Radio link aspects
- TDMA Format

• Features of GSM Technology

There are many features associated with GSM technology due to which it is bar far the most leading mobile communication technology in the world today. GSM technology facilitates with high speed integrated data, voice data, fax, mail, voice male and mostly used SMS feature. GSM also make sure that all the communication made between networks are secured and protected



from intruders and frauds

1. One of the major advantages of GSM technology which changed the way we looked mobile phones at the beginning. GSM actually brought the concept of being Mobile way beyond the limits. It enabled us to communicate across the continents.
2. GSM supports multiple frequency levels like 900 MHz, 1800 MHz, 1900 MHz. 1900MHz frequency is used in North America where as 1800MHz is used in other parts of the world. Different frequency bands are used by different mobile phone operators. If you are using Mobile phone which supports 1800MHz and operators are available on this band, phone can be used the network, where as if phone is out of the range of the frequency band on mo-



mobile operator frequency than you need to have phone that supports the frequency. To avoid such cases one should always adopt to have mobile phones that support multiple frequency bands.

- **Characteristics of GSM Standard**

- Fully digital system using 900,1800 MHz frequency band.
- TDMA over radio carriers(200 KHz carrier spacing.
- 8 full rate or 16 half rate TDMA channels per carrier.
- User/terminal authentication for fraud control.
- Encryption of speech and data transmission over the radio path.
- Full international roaming capability.



- Low speed data services (upto 9.6 Kb/s).
 - Compatibility with ISDN.
 - Support of Short Message Service (SMS).
- **Advantages of GSM over the Analog Systems**
 - It provides better quality and spectrum usage, increasing the number of subscribers.
 - A GSM subscriber may move with his phone and use it within all countries covered by GSM.
 - It can provide many new user services, such as data and Fax, SMS etc.
 - It introduces lower cost equipment such as powerful handheld mobile phones.

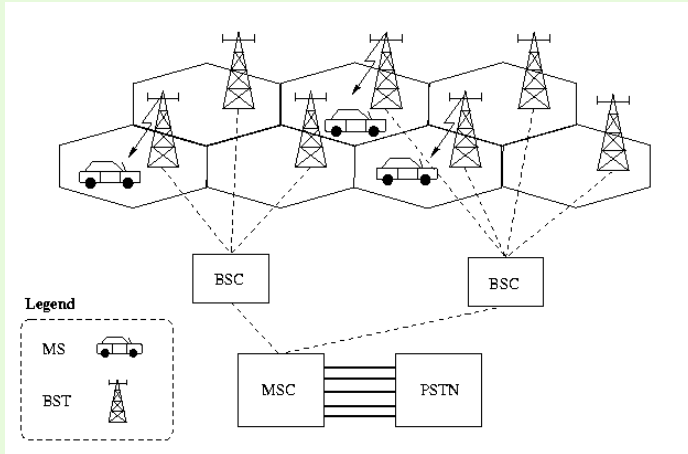


Fig. Architecture of GSM networks

GSM network is consist of thee major systems, it can also be considered architecture of GSM networks. These three systems



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are SS, which is known as **The Switching System**, BSS, which is **The Base Station** and the other important system is, **The operation and support System** for GSM networks.

Below all three systems are defined in details with sub systems of each system.

- **The Switching System**

The Switching system is a very operative system in which many crucial operations are conducted, SS systems holds five databases with in it which performs different functions. Its major tasks are that it performs call processing and subscriber related functions. These five databases from SS systems are HLR, MSC, VLR, AUC and EIR. Below each database is explained in detail where you will learn what functions this systems performs.

**i) HLR- Home Location Register:**

HLR is database, which holds very important information of subscribers. It is mostly known for storing and managing information of subscribers. It contains subscriber service profile, status of activities, information about locations and permanent data of all sorts. When new connections are purchased, these subscribers are registered in HLR of mobile phone companies.

ii) MSC- Mobile Services Switching Center:

The Mobile Services Switching Center (MSC) is at the center of the GSM system. It has direct access to the Base Station System and the relevant databases and other networks. It is built to perform switching functionality of the entire system. It's most important task is to control the calls to and from other telephones, which means it controls calls from same networks



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and calls from other networks. Toll ticketing, common channel signaling, network interfacing etc are other tasks which MSC is responsible for. MSC's other functions include:

- Setting up a call, routing, controlling and terminating a speech connection.
- Controlling location updating and handovers.
- Handling the subscriber's supplementary services.
- Collecting charging and accounting information.
- Handles technical end of telephony

iii) VLR- Visitor Location Register:

The Visitor Location Register (VLR) is integrated with the MSC to reduce the signaling between the two nodes. It is a database containing information about all mobile subscribers



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currently located in the service area of one MSC. Each MSC must therefore have its own unique VLR. The combined node is called MSC/VLR. It stores temporary data regarding subscribers which is needed by Mobile Services Switching Center- MSC VLR is directly connected to MSC, when subscriber moves to different MSC location, Visitor location register – VLR integrates to MSC of current location and requests the data about subscriber or Mobile station (MS) from the Home Location Register –HLR. When subscriber makes a call the Visitor location register - VLR will have required information for making call already and it will not required to connect to Home Register Location - HRL again.

iv) AUC- Authentication Center:



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AUC is a small unit which handles the security end of the system. Its major task is to authenticate and encrypt those parameters which verify user's identification and hence enables the confidentiality of each call made by subscriber. Authentication center – AUC makes sure mobile operators are safe from different frauds most likely to happen when hackers are looking for even smallest loop wholes in systems.

v) EIR – Equipment Identity Register:

EIR is another important database which holds crucial information regarding mobile equipments. EIR helps in restricting for calls been stolen, mal functioning of any MS, or unauthorized access. AUC – Authentication center and EIR- Equipment Identity registers are either Stand-alone nodes or some times work together as combined AUC/EIR nodes for optimum performance.



- **The Base Station System (BSS)**

The base station system has a very important role in mobile communication. BSS are basically out door units which consist of iron rods and are usually of high length. BSS are responsible for connecting subscribers (MS) to mobile networks. All the communication is made in Radio transmission. The Base station System is further divided in two systems. These two systems are BSC, and BTS.

- i) **BTS – The Base Transceiver Station:**

Subscriber, MS (Mobile Station) or mobile phone connects to mobile network through BTS; it handles communication using radio transmission with mobile station. As name suggests, Base transceiver Station is the radio equipment which receive and transmit voice data at the same time. BSC control group of



BTSs.

ii) BSC – The Base Station Controller:

The Base Station normally controls many cells, it registers subscribers, responsible for MS handovers etc. It creates physical link between subscriber (MS) and BTS, then manage and controls functions of it. It performs the function of high quality switch by handover over the MS to next BSC when MS goes out of the current range of BTS, it helps in connecting to next in range BTS to keep the connection alive within the network. It also performs functions like cell configuration data, control radio frequency in BTS. Data moves to MSC-Mobile switching center after BSC done processing it. MSC is switching center which acts as bridge between different mobile networks.



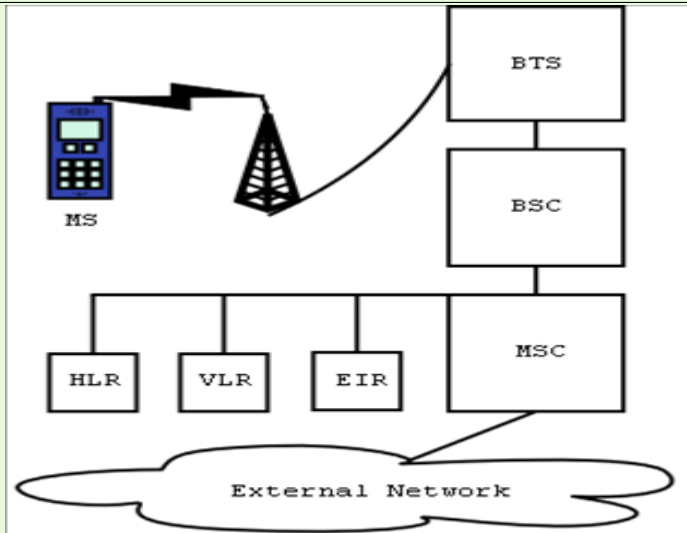


Fig. 6.1 How GSM works



- **The Operation and Support System (OSS)**

OMC- Operations and maintenance center is designed to connect to equipment of MSC- Mobile Switching Center and BSC- Base Station Controller. The implementation of OMC is called OSS-The Operations and Support System. OSS helps in mobile networks to monitor and control the complex systems. The basic reason for developing operation and support system is to provide customers a cost effective support and solutions. It helps in managing, centralizing, local and regional operational activities required for GMS networks.

Maintaining mobile network organization, provide overview of network, support and maintenance activities are other important aspects of Operation and Support System.



• Different Frequency Bands

There are three different frequency bands on which mobile phones are usually operates and these are Dual Band, Tri-Band and Quad Band.

1. **Dual Band:** Dual frequency band operates on 900MHz and 1800 MHz, that means mobile phone that supports dual band can be operated any where in the world where 900 MHz and 1800 MHz frequencies are used. Dual Band GSM networks usually found in all continents Europe, Asia, Africa, Australia and South America.
2. **Tri-Band:** As name is obvious three frequencies are supported in Tri Band, these frequencies are 900 MHz, 1800MHz and 1900 MHz. Tri band is also supported all around the world these days.



3. **Quad-Band:** Quad Band supports four frequencies which are 850 MHz, 900 MHz, 1800 MHz, 1900 MHz. Quad band also enables GSM phones to roam almost anywhere in the world. All countries support GSM networks hence make communication possible.

GMSC: Gateway Mobile Switching Center

This is an exchange that interconnects two networks, for example PSTN calls to a mobile subscriber and are routed to the GMSC. Gateway MSC (GMSC) is used to connect the GSM network with other networks. It is, in other words, an interface between the internal cellular network and PSTN.



- **Calling in a GSM network**

A mobile station places a call by sending a call initiation request to its nearest base station. This request is sent on a special channel, the reverse control channel (RCC). The base station sends the request, which contains the telephone number of the called party, to the MSC. The MSC validates the request and uses the number to make a connection to the called party via the PSTN. It first connects itself to the MSC of the called party, then the MSC instructs the base station and mobile station that placed the call to switch to voice channels. The mobile station that placed the call is then connected to the called station, using unused Forward and Backward Voice Channels (FVC, BVC).

The steps that take place when a mobile station receives an incoming call are as follows:



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- Mobile stations continually scan the forward control channel (FCC) for paging signals from base stations. When a MSC receives a request for a connection to a mobile station in its area, it sends a broadcast message to all base stations under its control. The message contains the number of the mobile station that is being called. The base stations then broadcast the message on all forward control channels (FCC). The correct mobile station acknowledges the page, by identifying itself over the reverse control channel (RCC). The MSC receives the acknowledgment via the base station, and instructs the base station and mobile station to switch to an unused voice channel. A data message is then transmitted over the forward voice channel, which instructs the mobile phone to ring.



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The steps explained above happen fast enough that the user does not experience any noticeable delay between placing a request for a call and the call being connected.

- **GSM Enhancements**

To enhance the current data capabilities of GSM, operators and infrastructure providers have specified new extensions to GSM Phase II. These extensions are:

- High Speed Circuit Switched Data (HSCSD): by using several circuit channels.
- General Packet Radio Service (GPRS) to provide packet radio access to external packet data networks (such as X.25 or Internet)
- Enhanced Data rate for GSM Evolution (EDGE): using



a new modulation scheme to provide up to three times higher throughput (for HSCSD and GPRS)

- Universal Mobile Telecommunication System (UMTS): a new wireless technology using new infrastructure deployment.

6.7.2. CDMA

CDMA stands for Code Division Multiple Access. In CDMA, the same bandwidth is occupied by all the users, however they are all assigned separate codes, which differentiates them from each other. CDMA utilizes a spread spectrum technique in which a spreading signal (which is uncorrelated to the signal and has a large bandwidth) is used to spread the narrow band message signal. Almost all accepted 3G radio standards are based



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on CDMA CDMA2000, W-CDMA and TD-SCDMA. Each cell is allocated a frequency bandwidth split into forward and reverse signals. Transmission is in form of direct sequence spread spectrum (DS-SS). DS-SS uses a chipping code (pseudo random number). CDMA precise power control of mobiles is critical in maximizing systems capacity and multiple access is provided by assigning orthogonal chipping codes to multiple users.

6.7.3. WCDMA (UMTS)

WCDMA is based on DS-CDMA (direct sequence code division multiple access) technology in which user-information bits are spread over a wide bandwidth (much larger than the information signal bandwidth) by multiplying the user data with the spreading code. The chip (symbol rate) rate of the spreading sequence



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is 3.84 Mcps, which, in the WCDMA system deployment is used together with the 5-MHz carrier spacing. The processing gain term refers to the relationship between the signal bandwidth and the information bandwidth. Thus, the name wideband is derived to differentiate it from the 2G CDMA (IS-95), which has a chip rate of 1.2288 Mcps. In a CDMA system, all users are active at the same time on the same frequency and are separated from each other with the use of user specific spreading codes.

The wide carrier bandwidth of WCDMA allows supporting high user-data rates and also has certain performance benefits, such as increased multi-path diversity. The actual carrier spacing to be used by the operator may vary on a 200-kHz grid between approximately 4.4 and 5 MHz, depending on spectrum arrangement and the interference situation.



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In WCDMA each user is allocated frames of 10 ms duration, during which the user-data rate is kept constant. However, the data rate among the users can change from frame to frame. This fast radio capacity allocation (or the limits for variation in the uplink) is controlled and coordinated by the Radio Resource Management (RRM) functions in the network to achieve optimum throughput for packet data services and to ensure sufficient Quality of Service (QoS) for circuit-switched users. WCDMA supports two basic modes of operation: FDD and TDD. In the FDD mode, separate 5-MHz carrier frequencies with duplex spacing are used for the uplink and downlink, respectively, whereas in TDD only one 5-MHz carrier is time shared between the uplink and the downlink. WCDMA uses coherent detection based on the pilot symbols and/or common



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pilot. WCDMA allows many performance - enhancement methods to be used, such as transmit diversity or advanced CDMA receiver concepts. Table summaries the main WCDMA parameters.

The support for handovers (HO) between GSM and WCDMA is part of the first standard version. This means that all multi-mode WCDMA/GSM terminals will support measurements from the one system while camped on the other one. This allows networks using both WCDMA and GSM to balance the load between the networks and base the HO on actual measurements from the terminals for different radio conditions in addition to other criteria available.



- **Advantages of CDMA**

- **Frequency diversity** – frequency-dependent transmission impairments have less effect on signal
- **Multipath resistance** – chipping codes used for CDMA exhibit low cross correlation and low auto correlation.
- **Coverage** - CDMA results in coverage that is between 1.7 and 3 times that of TDMA. Coding and interleaving provides the ability to cover a large area for the same amount of available power used in other systems
- **Privacy** – Privacy is inherent since spread spectrum is obtained by use of noise-like signals
- **Graceful degradation** – system only gradually degrades as more users access the system



- **Flexible network planning** (planning is no longer needed)
- **Disadvantages of CDMA**
 - **Self-jamming:** This is a steep deterioration of performance as a result of poor synchronization. Poor synchronization causes partial-correlation with the codes of other users and the result will be a vast increase of the interference
 - **Near-far problem:** signals closer to the receiver are received with less attenuation than signals farther away
 - **Soft handoff:** requires that the mobile acquires the new cell before it relinquishes the old; this is more complex than hard handoff used in FDMA and TDMA schemes.
 - **Throughput:** Low throughput efficiency for large num-



ber of users.

6.7.4. EDGE

EDGE stands for Enhanced Data rates for GSM Evolution. EDGE is most talked about technology in mobile communication technologies in recent times and it turn out to be the evolution of most widely used GSM technology. EDGE technology facilitates better data transmission rates with improved spectrum efficiency and the best part of EDGE is capability of supporting new applications and improved mobile communication capabilities. EDGE can also be known as extended version of GPRS therefore it is also called EGPRS.

EDGE is way to achieve better data rates on GSM networks. EGDE is add on to GPRS network and can not work stand



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alone. It works with new modulation techniques and new channel coding that is used for both Packet switch and circuit switch voice and data networks. Higher throughput and data rates can be achieved by adding new modulation and coding channels to already installed GPRS system.

EDGE is enhanced technology for 2G, 2.5 generation systems, GSM and GPRS networks. It makes deployment of EDGE technology that much easier for mobile companies. It does not require any up gradation in hardware, software or any changes in GSM network. EDGE supported transceiver are required to be installed, base station system are also required to be upgrade to support EDGE.

EDGE and GRPS technologies have similarities in it, these two technologies have same symbol rate, although modulation



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bit rates are different. EDGE technology can deliver three times faster bit rate ratio comparing to GPRS in given time, this is how EDGE transmit at better rate. GPRS can transfer data at rates of 115 kbps theoretically and up to 160 kbps on physical layer, where as EDGE/EGRPS can transfer up to 384 kbps on physical layer and 473.6 kbps theoretically. Latest modulation and error tolerant transmission methodologies with help of mechanism of improved link adaption have made rates of EGPRS/EDGE possible. This has enabled improved spectrum efficiency and usability of modern applications like Wireless internet, emailing, file transferring and video calling.



• Advantages of EDGE

- The good and practical aspect of implementing Edge technology is its adaptability to the existing system like GPRS, HSCSD (high speed circuit switched data) and specifications; it does not require new systems to be developed for deploying it.
- EDGE/EGPRS can be operated on existing frequency spectrum that is 800MHz, 1800 MHz and 1900 MHz.
- EGPRS/EDGE is cost effective when it comes installing it on the network, it requires small units of improvement and enhancements on existing network.
- The most obvious advantage of EDGE is it provides thrice as better speed as GPRS does.



- It improves performance of existing applications and room for new applications like video calls on mobile etc.

6.7.5. GPRS

GPRS stands for General Packet Radio Services. It is one of the technologies to improve 2G phones (second generation phones) to enable them for transferring data at higher speed, GPRS allows mobile phones to remain connected to network and transfer requested or sent data instantly, e.g. if you receive MMS from other mobile phone, you do not need to press a button to check if you have any new MMS, instead mobile handset notifies you when new MMS is downloaded to your mobile. GPRS technology can provide you up to 32 kbps to 48 kbps. One of the features of GPRS technology which makes it even more useful



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and practical is that data can transfer during the call and there is no requirement of disconnecting call to receiving incoming or out going data. For example, you can receive SMS during the call and you can reply to it without having to disconnect your on going call. GPRS usage is charged for the amount of data sent and received, unlike CSD which is charged for amount of time it remained connected.

GPRS technology enabled to move 2G networks closer to the performance of 3G networks. 3G (Third Generation) support networks are best at transferring data to unbelievably higher data rates which can go up 2 Mbps. Data rate up to 2Mbps made so many impossible tasks very possible on mobile phones which was once only possible on computers. With this data rate it is possible to do video calling, video streaming and download-



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ing, web browsing, emailing etc. Improved 2G networks (GPRS) manages to support few of applications like web browsing, emailing, video streaming, Multimedia messaging service etc. therefore GPRS can also be said that it is 2.5 generation technology. Technologies like HSCSD and EDGE are enhancements to the CSD technology

GRPS have numerous classes. Each class represents the amount of data that can be transferred with in the network. All classes have number timeslots which are available for uploading data to the phone or download data to the phone. Voice calls have additional times slots for communication and does not use timeslots dedicated for transferring of data. The more the timeslots are available more and faster the data can be transferred as these timeslots are used simultaneously. All timeslots



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are shared over the network by the users; GRPS send and receive data in packets and all the slots are not used all the time, hence increase the overall performance and capacity of the network. During the peak time, lower data rates can be experienced as network will give higher priority to the voice calls.

- **GPRS features**

1. Radio resources are allocated for only one or a few packets at a time, so GPRS enables:
 - Higher data rates than gsm up to 164kbps
 - many users to share radio resources, and allow efficient transport of packets
 - connectivity to external packet data networks
 - volume-based charging



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- The Capital expenditure (capex) and operational expenditure (opex) are minimized due to the upgrade of the existing infrastructure.
2. GPRS also carries SMS in data channels rather than signaling channels as in GSM
- **GPRS components**
 - PCU: The PCU is a logical network element that is responsible for air-interface access control, packet scheduling on the air interface, and packet assembly and reassembly.
 - SGSN: Is analogous to the MSC in the circuit-domain, but performing in a packet domain, including mobility management, security, and access control functions.
 - Gateway GPRS support node (GGSN): The Gateway GPRS



Support Node acts as an interface and a router to external networks. The GGSN contains routing information for GPRS mobiles, which is used to tunnel packets through the IP based internal backbone to the correct Serving GPRS Support Node. The GGSN also collects charging information connected to the use of the external data networks and can act as a packet filter for incoming traffic.

6.7.6. Third generation (3G)

3G is a generation of mobile phone standards and technology, after 2G and it is based on the International Telecommunication Union (ITU) family of standards under the International Mobile Telecommunications programme, "IMT-2000". Third generation (3G) networks were conceived from the Universal Mo-



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Mobile Telecommunications Service (UMTS) concept for high speed networks for enabling a variety of data intensive applications. 3G systems consist of the two main standards, CDMA2000 and W-CDMA, as well as other 3G variants such as NTT DoCoMo's Freedom of Mobile Multimedia Access (FOMA) and Time Division Synchronous Code Division Multiple Access (TD-SCDMA) used primarily in China. This technology enables network operators to offer users a wider range of more advanced services while achieving greater network capacity through improved spectral efficiency. It provides high speed wireless communications to support multimedia, data, and video in addition to voice.

- **Objectives of 3G**

- Use of common global frequencies for all cellular networks.



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- Worldwide roaming.
- Standardization of radio interfaces.
- High data transmission rates for both circuit and packet switched data -Data rate -384 kbps while moving and 2mbps for stationary
- Efficient spectrum utilization schemes
- Symmetrical / asymmetrical data transmission rates

• Data Speed

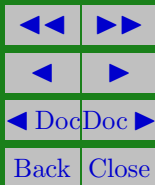
The data speed of 3G is determined based on a combination of factors including the chip rate, channel structure, power control, and synchronization. Actual data speeds vary in accordance with several factors including:



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- Number of users in cell/sector
- Distance of user from cell
- User is moving or stationary
- Network operator capacity and network optimization requirements

1xEV-DO is a data-only solution, supporting a theoretical data speed of up to 2.457 Mbps 1xEV-DV is a data and voice solution, supporting a theoretical data speed of up to 3.072 Mbps FOMA has two operational modes, supporting a dedicated 64 Kbps connection or a 384 Kbps downlink/64 Kbps uplink best-effort connection. TD-SCDMA can operate in 1.6 MHz or 5 MHz mode for 2 Mbps or 6 Mbps respectively.





- **The Drive for 3G**

The major drive for 3G is to provide faster data speed for data-intensive applications such as video. In addition, 3G to providing faster data speeds on a per-user basis, 3G is also helpful to provide greater overall capacity for voice and data users. For example, NTT DoCoMo's plan to migrate iMode users from the 2G PDC network to FOM is driven by overall capacity concerns, as apposed to individual user data speed requirement.

- **3G features**

- The most significant feature of 3G mobile technology is that it supports greater numbers of voice and data customers — especially in urban areas — and higher data rates at lower incremental cost than 2G.



- 3G uses .6 MHz channel carrier width to deliver significantly higher data rates and increased capacity compared to 2G networks.

• 3G Architecture

W-CDMA uses the same CN as GPRS, utilizing existing infrastructure such as the GGSN and SGSN. W-CDMA, however, does require new RAN infrastructure such as the Node B, which replaces the BTS, and the Radio Network Controller (RNC), which replaces the BSC. Ultimately, the W-CDMA CN will evolve to comprise a full Mobile IP infrastructure including Media Gateway (MGW) and Media Gateway Controller (MGC) equipment for VoIP and other new equipment such as the HSS and CSDF.



CDMA2000 starts with new channel cards and then migrates to a full Mobile IP infrastructure requiring new Core Network (CN) infrastructure such as the AAA server and Packet Data Server Node (PDSN).

- **The need for test access in wireless 3G**

Designing test access into emerging 3G networks represents the best way to ensure rapid deployment, minimize incident response time and gain efficiency in testing. As wireless networks evolve into multi-vendor, super-regional entities, new testing methods must be used to ensure standards compliance, network performance and uninterrupted service.

The service demands of 3G networks include more sophisticated content, more demanding users and, ultimately, fulfillment



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of service level agreements (SLAs), which have become common in non-wireless networks. To support these goals, a test access approach must make test equipment deployment:

- easy for any authorized user to access
- comprehensive, to meet all testing needs
- secure, to ensure control of the test environment
- vendor-independent, to ensure that any type of hardware or software can be used for testing
- technology-agnostic, to assure common procedures are used in whatever type of network is deployed
- scalable, to ensure response levels as the network grows in scope as well as speed
- rapid, to maximize responsiveness and minimize downtime



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By designing test access into its 3G network architecture, a network operator carrier can ensure that efficient testing and trouble resolution will be provided during evaluation, optimization and at launch. Furthermore, by considering NTA at the time the network is defined, testing can be automated and included in legacy OSS (Operational Support System) implementations.

This inclusion allows network operation centers to gracefully incorporate the new networks into their operations, minimizing training requirements and allowing broader testing availability. Finally, including NTA in the initial design of a network, or network expansion, eliminates service interruptions, which may be caused by retrofitting test access into a live network.



6.8. The Current and Future Technologies

The universal mobile telecommunications system (UMTS) proposes to achieve this ideal of PCS data, which includes services that are independent of location, network, or terminal. This system is hailed as being the next generation cellular system, and is referred to as 3G Cellular. In order to attain this ideal, all interest holders of the cellular world will have to work together to standardize communications. The road ahead for other cellular radio network architectures will be very similar to that of GSM.

6.8.1. High-Speed Circuit-Switched Data Service - HSCSD

The current limitation of 9.6 kbps over the radio link is due to the fact that only one of the eight possible TDMA channels is used for data. This limitation has been alleviated slightly by



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using less error checking, which raises the transmission rate to 14.4 kbps in the parts of the network that can afford to use less error checking.

The obvious way to increase transmission rates is to use more than one channel for data transmission. Currently being tested is a service called high-speed circuit-switched data service (HSCSD). This service combines two to four of the channels for data transmission to attain rates of 28.8 and 56 kbps. Such data rates open a whole new world to cellular radio networks. Users will be able to connect to the Internet and their own dial up network servers through the GSM network. The GSM network provides encryption as a standard service, therefore the implementation of Virtual Private Networks on GSM networks is a natural extension of data services already provided.



6.8.2. 4G technology

A descendant to 2G and 3G aiming to provide the very high data transfer rates. This technology can provide very speedy wireless internet access to not only stationary users but also to the mobile users. This technology is expected to trounce the deficiencies of 3G technology in terms of speed and quality. 4G can be best described in one word “MAGIC”, which stands for Mobile multimedia Anytime Anywhere Global mobility support, integrated wireless and personalized services.

4G is a multi purpose and versatile technology hence it can utilize almost all of the packet switched technologies. It can use both orthogonal frequency division multiplexing (OFDM) and orthogonal frequency division multiple access (OFDMA). OFDM mechanism splits a digital signal into different narrow-



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band and frequencies. The reason why 4G makes use of this technology lies in its ability to minimize the intervention among symbols and channels associated to data streaming. 4G is also capable of using multiple input / multiple output technology (MIMO). This antenna technology is used to optimize the data speed and reduce the errors in the networks. Universal Mobile Telecommunication Service (UMTS) which is basically a broadband 3G technology is also a part of 4G. This broadband technology transfers data in the form of frames or packets. Hence it is capable of carrying voice, video, text and other types of multimedia datagram with the speed of 2Mb. UMTS is part of 4G because it can enable 4G to make use of international mobile phone roaming via using GSM (Global system for Mobile Communications). Another wireless telecommunication technology



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ogy known as time division synchronous code division multiple access (TD-SCDMA) provides support to 4G to transfer both circuit switched data like video and voice and packet switched data.

6.9. Fading

The term fading, or, small-scale fading, means rapid fluctuations of the amplitudes, phases, or multipath delays of a radio signal over a short period or short travel distance. This might be so severe that large scale radio propagation loss effects might be ignored.

6.9.1. Multipath Fading Effects

In principle, the following are the main multipath effects:



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1. Rapid changes in signal strength over a small travel distance or time interval.
2. Random frequency modulation due to varying Doppler shifts on different multipath signals.
3. Time dispersion or echoes caused by multipath propagation delays.

6.9.2. Factors Influencing Fading

The following physical factors influence small-scale fading in the radio propagation channel:

1. Multipath propagation: Multipath is the propagation phenomenon that results in radio signals reaching the receiving antenna by two or more paths. The effects of multipath include constructive and destructive interference, and



phase shifting of the signal.

2. Speed of the mobile: The relative motion between the base station and the mobile results in random frequency modulation due to different doppler shifts on each of the multipath components.
3. Speed of surrounding objects: If objects in the radio channel are in motion, they induce a time varying Doppler shift on multipath components. If the surrounding objects move at a greater rate than the mobile, then this effect dominates fading.
4. Transmission Bandwidth of the signal: If the transmitted radio signal bandwidth is greater than the "bandwidth" of the multipath channel (quantified by coherence bandwidth), the received signal will be distorted.



Revision Questions

EXERCISE 1.  Consider two mobiles belonging to the same home network in a GSM system. Explain step by step the process in GSM roaming, when one mobile wakes up in a foreign network and makes a call to the other.

Example . Briefly, state and explain the frequency bands

Solution:

Dual Band: Dual frequency band operates on 900MHz and 1800 MHz, that means mobile phone that supports dual band can be operated any where in the world where 900 MHz and 1800 MHz frequencies are used. Dual Band GSM networks usually found in all continents Europe, Asia, Africa, Australia and South America.



Tri-Band: As name is obvious three frequencies are supported in Tri Band, these frequencies are 900 MHz, 1800MHz and 1900 MHz. Tri band is also supported all around the world these days.

Quad-Band: Quad Band supports four frequencies which are 850 MHz, 900 MHz, 1800 MHz , 1900 MHz. Quad band also enables GSM phones to road almost anywhere in the world. All countries support GSM networks hence make communication possible. ☐

EXERCISE 2.  How does EDGE technology differ from GPRS ?



Solutions to Exercises

Exercise 1.

When the mobile device is turned on or is transferred via a handover to the network, this new "visited" network sees the device, notices that it is not registered with its own system, and attempts to identify its home network. If there is no roaming agreement between the two networks, maintenance of service is impossible, and service is denied by the visited network.

The visited network contacts the home network and requests service information (including whether or not the mobile should be allowed to roam) about the roaming device using the IMSI number.

If successful, the visited network begins to maintain a tem-



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porary subscriber record for the device. Likewise, the home network updates its information to indicate that the mobile is on the host network so that any information sent to that device can be correctly routed.

Exercise 1



Exercise 2.

EDGE technology can deliver three times faster bit rate ratio comparing to GPRS in any given time.

EDGE allows the delivery of advanced mobile services such as the downloading of video and music clips, full multimedia messaging, high-speed colour Internet access and e-mail on the move.

Using EDGE, operators can handle three times more subscribers than GPRS; triple their data rate per subscriber, or add extra capacity to their voice communications.

EDGE is a digital mobile phone technology but GPRS is a mobile data service.

EDGE allows existing TDMA and GSM carriers to provide



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3G services whereas GPRS allows both 2G and 3G communication systems.

GPRS can transfer data at rates of 115 kbps theoretically and up to 160 kbps on physical layer, where as EDGE/EGRPS can transfer up to 384 kbps on physical layer and 473.6 kbps theoretically.

Exercise 2